understanding of gravity, whose influence becomes weaker as the distance between objects becomes larger.

An important point to note is that the ball bearing itself warps the rubber membrane, although only slightly. Similarly, the earth, being a massive body in its own right, also warps the fabric of space, although far less than the sun. This is how, in the language of general relativity, the earth keeps the moon in orbit, and it is also how the earth keeps each of us bound to its surface. As a skydiver plunges earthward, he or she is sliding down a depression in the spatial fabric caused by the earth's mass. Moreover, each of us—like any massive object—also warps the spatial fabric in close proximity to our bodies, although the comparatively small mass of a human body makes this a minuscule indentation.

In summary then, Einstein fully agreed with Newton's statement that "Gravity must be caused by an agent" and rose to Newton's challenge in which the identity of the agent was left "to the consideration of my readers." The agent of gravity, according to Einstein, is the fabric of the cosmos.

A Few Caveats

The rubber membrane-bowling ball analogy is valuable because it gives us a visual image with which we can grasp tangibly what we mean by a warp in the spatial fabric of the universe. Physicists often use this and similar analogies to guide their own intuition regarding gravitation and curvature However, its usefulness notwithstanding, the rubber membrane-bowling ball analogy is not perfect and for clarity we call attention to a few of its shortcomings.

First, when the sun causes the fabric of space around it to warp this is not due to its "being pulled downward" by gravity as in the case of the bowling ball, which warps the rubber membrane because it is pulled earthward by gravity. In the case of the sun, there is no other object to "do the pulling." Instead, Einstein has taught us that the warping of space is gravity. The mere presence of an object with mass causes space to respond by warping. Similarly, the earth is not kept in orbit because the gravitational pull of some other external object guides it along the valleys in the warped spatial environment, as occurs for a ball bearing on the warped rubber membrane. Instead, Einstein showed that objects move through space (spacetime, more precisely)